

Lab 9 - Numerical Integration

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Problem 1 Answer

```
f = @(x) exp(x).*cos(x);
xs = [];
h = 0;
CompArea = [];
E = [];
Er = [1];
TrueArea = (1/2)*(exp(pi/2)-1);

for i=[1:8]
    xs = 0:pi/(2^(1+i)):pi/2;
    h = pi/(2^(1+i));
    CompArea = [CompArea; MyTrapezoidQuad(f,xs,h)];
    E = [E; abs(CompArea(i)-TrueArea)];

end

Er = [Er; E(2:end)./E(1:end-1)];
n = [2; 4; 8; 16; 32; 64; 128; 256]
t = table(n,CompArea,E,Er)

function [result] = MyTrapezoidQuad(f,xs,h)
    I = (h/2) * ((2*sum(f(xs))) - f(xs(1)) - f(xs(end)));
    result = I;
end
```

Output:

8×4 table

n	CompArea	E	Er
2	1.6108	0.29448	1
4	1.8308	0.074416	0.2527
8	1.8866	0.018652	0.25064
16	1.9006	0.0046659	0.25016
32	1.9041	0.0011667	0.25004
64	1.9049	0.00029168	0.25001
128	1.9052	7.292e-05	0.25
256	1.9052	1.823e-05	0.25

Problem 2 Answer

```
f = @(x) exp(x).*cos(x);
xs = [];
```

```

h = 0;
CompArea = [];
E = [];
Er = [1];
TrueArea = (1/2)*(exp(pi/2)-1);

for i=[1:8]
    xs = 0:pi/(2^(1+i)):pi/2;
    h = pi/(2^(1+i));
    CompArea = [CompArea; MySimpsonQuad(f,xs,h)];
    E = [E; abs(CompArea(i)-TrueArea)];

end

Er = [Er; E(2:end)./E(1:end-1)];
n = [2; 4; 8; 16; 32; 64; 128; 256]
t = table(n,CompArea,E,Er)

function [result] = MySimpsonQuad(f,xs,h)
    I = (h/3) * ((2*sum(f(xs))) + (2*sum(f(xs(2:2:end-1)))) - f(xs(1)) -
f(xs(end)));
    result = I;
end

```

Output:

8×4 table

n	CompArea	E	Er
2	1.8859	0.019358	1
4	1.9042	0.001062	0.05486
8	1.9052	6.3806e-05	0.060081
16	1.9052	3.9469e-06	0.061857
32	1.9052	2.4603e-07	0.062337
64	1.9052	1.5367e-08	0.062459
128	1.9052	9.6029e-10	0.06249
256	1.9052	6.0016e-11	0.062498

Problem 3 Answer

```

format long;

f = @(x) exp(x).*cos(x);
xs = [];
h = 0;
CompArea = [];

a = 0;
b = pi/2;

```

```

for i=[1:8]
    xs = a:(b-a)/(2^i):b;
    h = (b-a)/(2^i);
    CompArea = [CompArea; MyTrapezoidQuad(f,xs,h)];
end

RichTotal = {};
RichCurrent = [];
RichPast = [];

for i=[1:7]
    for i2 = [1:8-i]
        if i == 1
            RichCurrent = [RichCurrent; ((2^(i+1))*CompArea(i2+1) - CompArea(i2))/
((2^(i+1))-1)];
        else
            RichCurrent = [RichCurrent; ((2^(i+1))*RichPast(i2+1) - RichPast(i2))/
((2^(i+1))-1)];
        end
    end
    RichCurrent = [RichCurrent; ones(i,1)]
    RichTotal{end+1} = RichCurrent;
    RichPast = RichCurrent;
    RichCurrent = [];
end

n = [2; 4; 8; 16; 32; 64; 128; 256]
t = table(n,CompArea,RichTotal{1:7})

function [result] = MyTrapezoidQuad(f,xs,h)
    I = (h/2) * ((2*sum(f(xs))) - f(xs(1)) - f(xs(end)));
    result = I;
end

```

Output:

```

8x9 table

    n          CompArea          Var3          Var4
Var5          Var6          Var7          Var8
Var9

    _____  _____  _____  _____

    2    1.61075989620211    1.90417669298792    1.90531748335373
1.90523834902413    1.90523869585689    1.9052386904864    1.90523869048266
1.90523869048268

```

4	1.83082249379147	1.905174884558	1.90524329491973
1.90523868501837	1.90523869057031	1.90523869048269	
1.90523869048268	1		
8	1.88658678686637	1.90523474362451	1.9052389731372
1.90523869039681	1.90523869048406	1.90523869048268	
1	1		
16	1.90057275443498	1.90523844444812	1.90523870806809
1.90523869048133	1.9052386904827	1	
1	1		
32	1.90407202194483	1.90523867511559	1.9052386915805
1.90523869048265	1	1	
1	1		
64	1.9049470118229	1.90523868952239	
1.90523869055127	1	1	
1	1	1	
128	1.90516577009752	1.90523869042266	
1	1	1	1
1	1		
256	1.90522046034137	1	
1	1	1	1
1	1		

Problem 4 Answer

First table is Trapezoidal, second is Simpson's

% Integral a

n	CompArea	E	Er
2	0.32875	0.056018	1
4	0.28666	0.013929	0.24866
8	0.2762	0.0034757	0.24952
16	0.2736	0.00086834	0.24983
32	0.27294	0.00021704	0.24994
64	0.27278	5.4255e-05	0.24998
128	0.27274	1.3564e-05	0.24999
256	0.27273	3.3909e-06	0.25

n	CompArea	E	Er
2	0.27166	0.0010672	1
4	0.27263	0.00010003	0.093733
8	0.27272	8.9088e-06	0.08906
16	0.27273	7.666e-07	0.08605
32	0.27273	6.4443e-08	0.084063
64	0.27273	5.3295e-09	0.082702
128	0.27273	4.3561e-10	0.081736

256	0.27273	3.5299e-11	0.081032
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% Integral b

n	CompArea	E	Er
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2	2.1667	0.17311	1
4	2.2687	0.071099	0.41071
8	2.3323	0.0074958	0.10543
16	2.3378	0.0019534	0.2606
32	2.3393	0.00048916	0.25041
64	2.3396	0.00012234	0.2501
128	2.3397	3.0588e-05	0.25003
256	2.3398	7.6473e-06	0.25001

n	CompArea	E	Er
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2	2.6251	0.28533	1
4	2.3027	0.037094	0.13001
8	2.3535	0.013705	0.36947
16	2.3397	0.00010593	0.0077293
32	2.3398	1.08e-06	0.010195
64	2.3398	6.7432e-08	0.062438
128	2.3398	4.2169e-09	0.062536
256	2.3398	2.6359e-10	0.062509

% Integral c

n	CompArea	E	Er
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2	6.2832	1.6717	1
4	7.9893	0.034397	0.020576
8	7.9549	1.2517e-06	3.639e-05
16	7.9549	7.9936e-15	6.3863e-09
32	7.9549	3.5527e-15	0.44444
64	7.9549	5.3291e-15	1.5
128	7.9549	3.5527e-15	0.66667
256	7.9549	5.3291e-15	1.5

n	CompArea	E	Er
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2	6.2832	1.6717	1
4	8.558	0.60311	0.36077
8	7.9435	0.011464	0.019008
16	7.9549	4.1723e-07	3.6395e-05
32	7.9549	2.6645e-15	6.3863e-09
64	7.9549	5.3291e-15	2
128	7.9549	2.6645e-15	0.5
256	7.9549	5.3291e-15	2

% Integral d

n	CompArea	E	Er
2	1.9212	0.018556	1
4	1.9351	0.0046152	0.24872
8	1.9386	0.0011523	0.24968
16	1.9394	0.00028799	0.24992
32	1.9397	7.1993e-05	0.24998
64	1.9397	1.7998e-05	0.25
128	1.9397	4.4994e-06	0.25
256	1.9397	1.1249e-06	0.25

n	CompArea	E	Er
2	1.9403	0.00053548	1
4	1.9398	3.1639e-05	0.059084
8	1.9397	1.9493e-06	0.061612
16	1.9397	1.214e-07	0.062278
32	1.9397	7.5808e-09	0.062444
64	1.9397	4.7369e-10	0.062486
128	1.9397	2.9603e-11	0.062495
256	1.9397	1.8496e-12	0.06248

% Integral e

n	CompArea	E	Er
2	0.56498	0.03502	1
4	0.58807	0.011927	0.34058
8	0.59602	0.0039819	0.33386
16	0.59869	0.0013106	0.32914
32	0.59957	0.00042692	0.32574
64	0.59986	0.000138	0.32324
128	0.59996	4.4347e-05	0.32137
256	0.59999	1.4189e-05	0.31995

n	CompArea	E	Er
2	0.58664	0.01336	1
4	0.59577	0.0042292	0.31657
8	0.59867	0.0013336	0.31534
16	0.59958	0.00042017	0.31506
32	0.59987	0.00013235	0.315
64	0.59996	4.1689e-05	0.31498
128	0.59999	1.3131e-05	0.31498
256	0.6	4.1361e-06	0.31498

% Integral f

n	CompArea	E	Er
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2	0.3125	0.0625	1
4	0.26562	0.015625	0.25
8	0.25391	0.0039062	0.25
16	0.25098	0.00097656	0.25
32	0.25024	0.00024414	0.25
64	0.25006	6.1035e-05	0.25
128	0.25002	1.5259e-05	0.25
256	0.25	3.8147e-06	0.25

n	CompArea	E	Er
2	0.25	0	1
4	0.25	0	NaN
8	0.25	0	NaN
16	0.25	0	NaN
32	0.25	0	NaN
64	0.25	0	NaN
128	0.25	0	NaN
256	0.25	0	NaN

There's a few interesting things to note here. First, integral a has Simpson's underperforming, this is likely because the third derivative has a jump discontinuity within our interval. This is also why it seems that Trapezoidal is working just fine, since it's only order 2 convergence, so the third derivative wouldn't effect it like it did for the order 4 of Simpson's. On the other hand, we see the same behaviour in integral e, since now the discontinuity is in the first derivative, effecting both methods. Integral c has incredible convergence (the weird error ratios at the end are due to floating point percision errors), and this is due to its periodicity, a fact that can be obviously seen when integral d doesn't have the same benefit due to the restricted interval. Finally, integral f has Trapezoidal work as normal, but Simpson's converges to the true value instantly with 100% accuracy. This is because Simpson's has degree 3 percision, and so for any polynomial of degree 3 or less, it has an error of 0% after just one iteration. This is unlike Trapezoidal which has degree 1 percision, and thus does not converge with 0% error in one step.

All of this converges as expected to theory when accounting for the specific nature of each function.